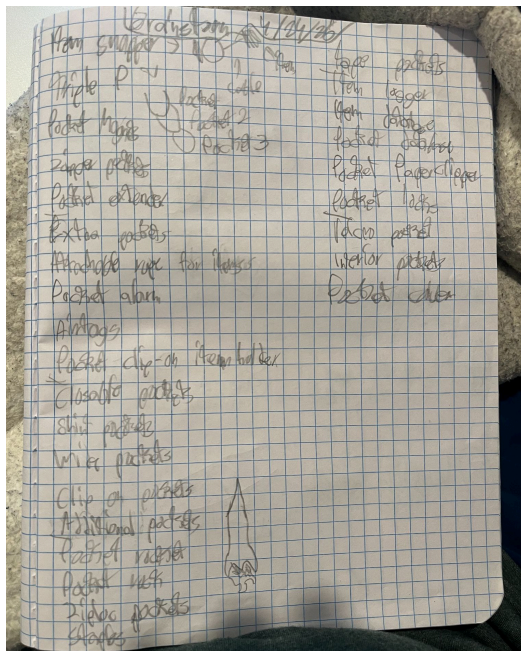


Element D: Design Concept Generation, Analysis, and Selection

People can easily lose things such as wallets, identification cards, phones, etc., from their pockets and need a way to keep things in their pockets because those items fall out of their pockets easily. Anyone who has had items fall out of their pockets would be involved in our project. This problem occurs everywhere in the world, from at home to traversing a big city. People often lose items from their pockets because their pockets are too small, so the items do not fully fit into the pockets.

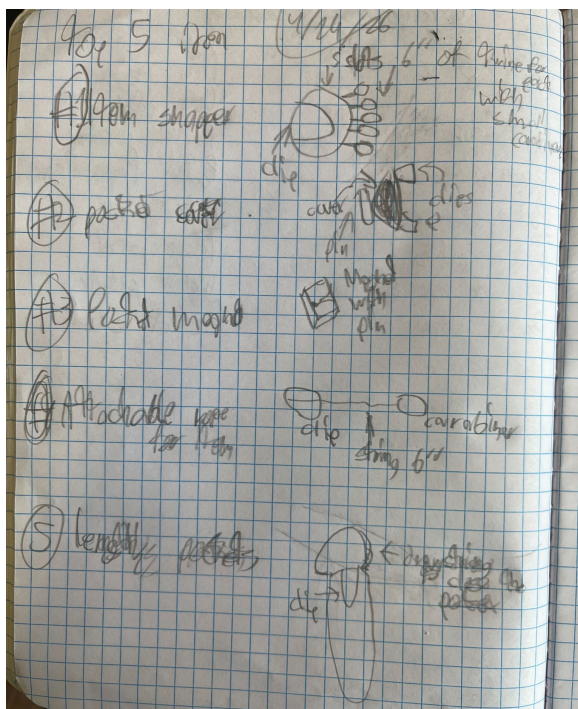
Our team's process for generating a solution starts with brainstorming. This portion is coming up with ideas that mostly pertain to what our problem is and could possibly help solve it. Notably, at this stage, we do not criticize or praise any singular solution. We only add onto them, giving our interpretation of it and more information about them. We came out with about 20 to 30 ideas for this.



Sheet with brainstorming ideas.

Next, we rule out the bad ideas and choose our top five ideas, meaning the five that we think would create the best solution. We also include very basic sketches of the

designs, just to get the bare minimum point across for later reference to choose our final solution. For us, this came down to the item snapper, pocket cover, pocket magnets, attachable ropes, and long pockets. Item snapper is a machine with a spring inside of it with a string connected to it so if you pull the string and release it, the string comes back, and the string has a carabiner attached to it. Pocket cover is a flap of fabric with pins that you can pin to your pockets to essentially put a cover over them. Pocket magnets are magnets with pins attached to the back of them so you can close your pockets with magnets. Attachable ropes are clips with strings attached to them, with carabiners attached to the other end, that can be clipped onto your pants. Long pockets are an external bag with drawstrings that can be clipped to your pockets.



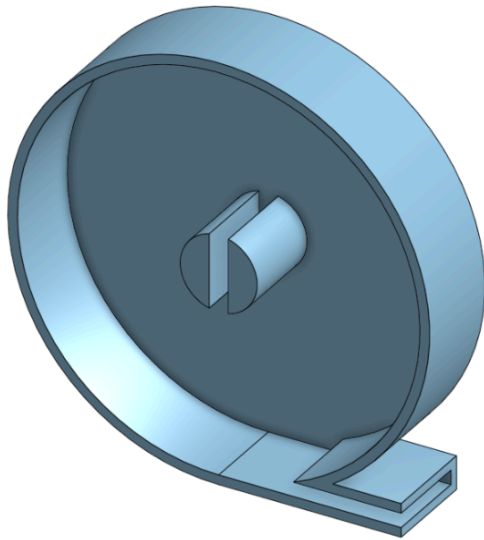
Sheet with the top five ideas.

Lastly, we used a decision matrix, which is a table that evaluates how well our top five ideas compare to the current solution to our problem while referencing our design requirements. This also includes weighting our design requirements, so our design

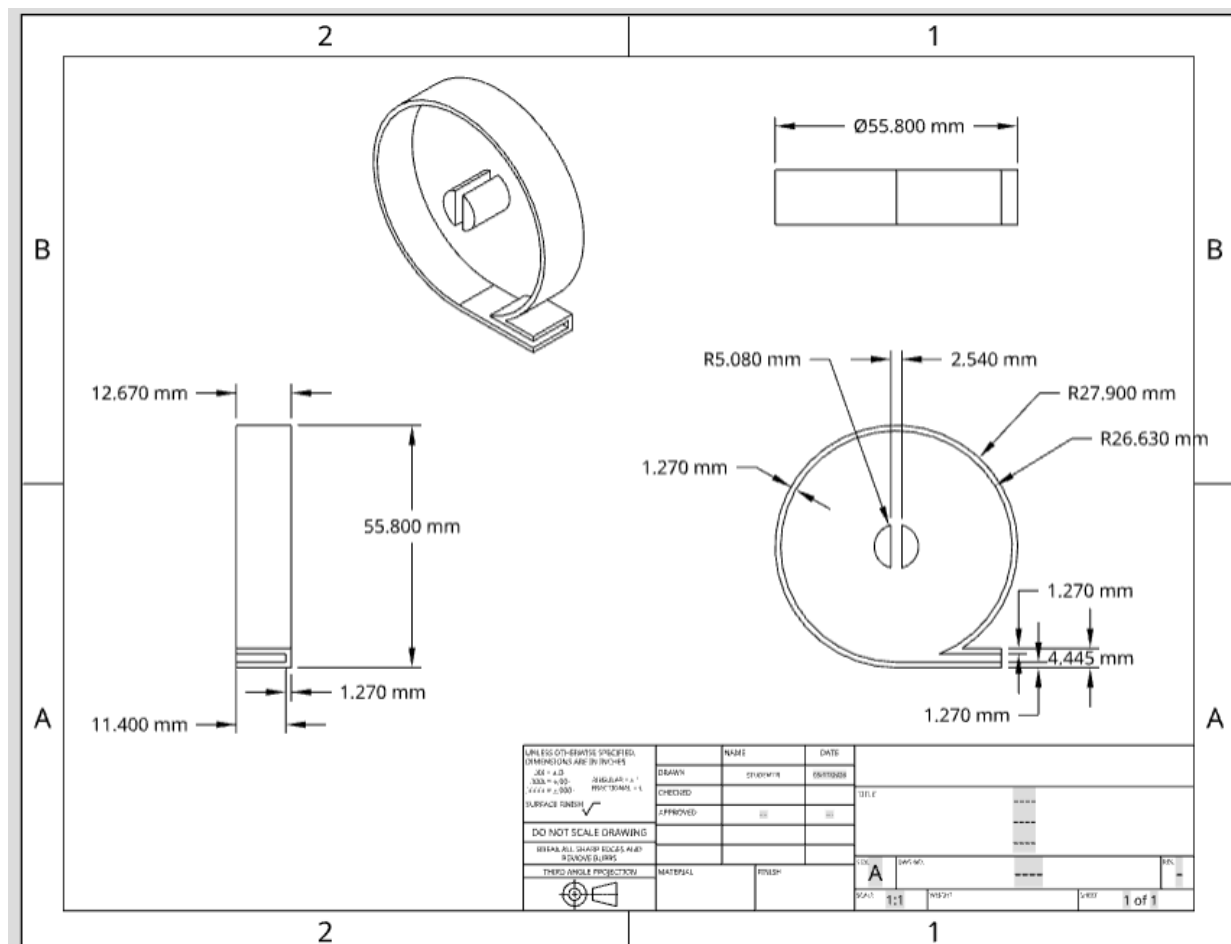
requirements of higher priority hold more value in the total outcome than our lower ones. In the end, this produced our final solution and the solution that our team went with, which was the item snapper.

	Proposed Solutions					
Design Requirements	Current Solution: Multi-pockets	Idea #1: Item snapper	Idea #2: Pocket cover	Idea #3: Pocket magnet	Idea #4: Attachable rope for item	Idea #5: Long pocket
Physical product x3	0	3	3	3	3	3
Safe product x3	0	-3	-3	-3	0	-3
Longevity x3	0	3	-3	3	0	-3
Under \$15 x3	0	-3	-3	-3	-3	-3
Does not interfere with electronics x3	0	0	0	-3	0	0
Lightweight x3	0	-3	-3	-3	-3	-3
Convenient size x2	0	2	2	2	2	-2
Removable x1	0	1	1	1	1	1
Waterproof x1	0	1	1	1	0	-1
Color options x1	0	1	1	1	1	1
Total Score	0	2	-4	-1	1	-10

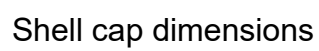
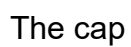
Our final design is an item that can be clipped to your pockets that has a spring and five strings attached to it. It resembles a badge clip. The spring is inside a 3D printed shell printed using 3D printer filament. The five strings are a stretchy bracelet string. Our design is 55.8mm x 12.67mm x 55.8mm

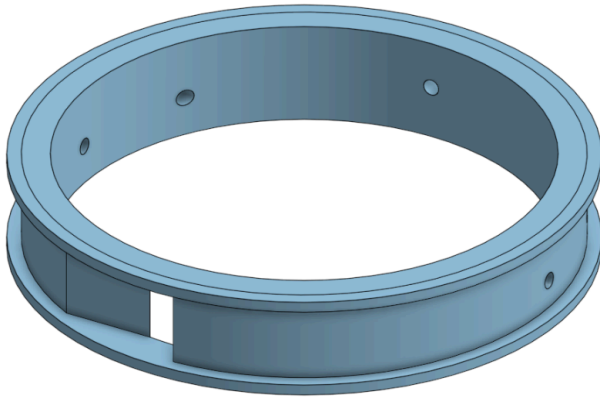


The shell

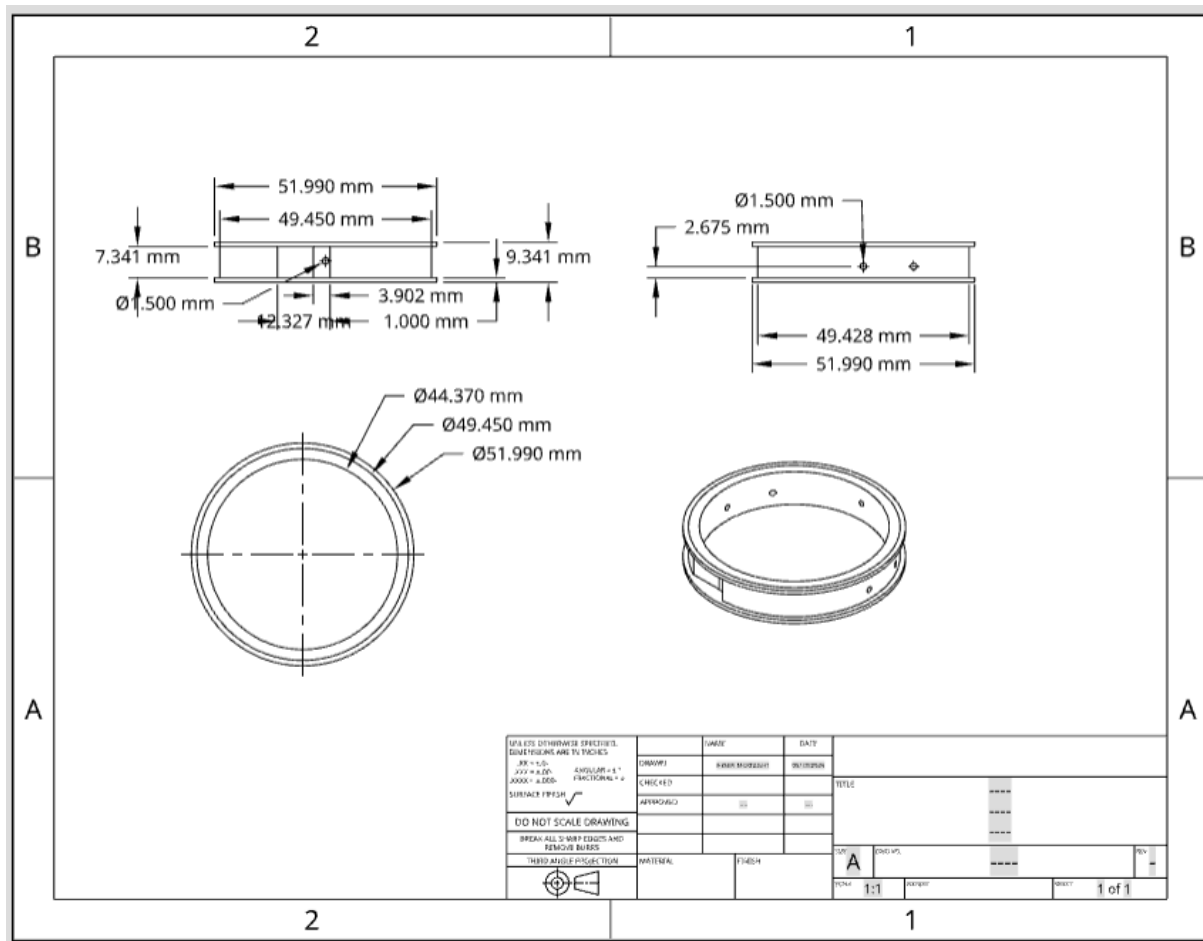


Shell dimensions





Spring holder



Spring holder dimensions

The spring for our design goes inside of the spring holder piece, and the hook for it goes on the rectangular cut out of it. The inside diameter of the interior piece is 44.37mm, and the total size is 51.99mm x 51.99mm x 9.341mm. Additionally, there are

five holes coming out of the spring holder which is where the string threads through.

The cap is 55.8 mm x 55.8 mm x 6.27 mm. It has a clip on one side. The clip's size is 31mm x 5mm x 15mm.

The shell is 55.8mm x 55.8mm x 12.67mm. Inside of it, there are two prongs. The wall size is 1.27mm. The inner diameter is 53.26mm. The depth of the middle area is 11.4mm.